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2D Finite Element Analysis of a Microstrip Transmission Line

MTH321 Project 2

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October 2026

Abstract

A finite-element capacitance is only useful if the distance to its limit is known. We compute the capacitance of a 3.0 mm-wide, 0.035 mm-thick trace on a 1.6 mm FR-4 substrate with conforming triangular P_1 elements, and we spend most of the report establishing that limit rather than the number.

The solver is pinned down before anything is measured. 23 analytic configurations are reproduced to roundoff, a manufactured solution returns the expected orders, and a layered interface keeps its normal flux continuous to 2.3×10^{-13}, so any remaining error is discretisation. Two independent routes, Richardson extrapolation of the project's own P_1 sequence and a separately written scikit-fem P_2 solve, then place the limit at $C_{\mathrm{ref}} = 120.467\,\mathrm{pF/m}$; they agree to 2.9×10^{-6} relative.

Against that limit the usual convergence indicator is misleading. At M7 the change from M6 is 0.156%, while the actual distance from the limit is 0.127%; the two differ by the factor $2^p - 1 = 1.23$, which a shrinking difference conceals unless p is known. The measured order is $p = 1.16$ for P_1 and 1.15 for P_2: doubling the polynomial degree does not buy an order, because the rate is set by the conductor corner and not by the basis.

Corner grading restores the order. Seven rounds of geometric refinement on the M5 background reach 0.017% with 139,426 unknowns, which is 7.6× closer than M7 at 39% of its degrees of freedom. The truncation of the $8H$ box contributes 0.18%, the same size as the discretisation error, and the algebraic error sits thirteen orders of magnitude below both, so the budget belongs entirely to the corner.

The submitted values are $C = 120.620\,\mathrm{pF/m}$, which is $0.153\,\mathrm{pF/m}$ or 0.127% from the extrapolated limit, and $C_{\mathrm{air}} = 35.968\,\mathrm{pF/m}$ on exactly the same M7 mesh, giving $\varepsilon_{\mathrm{eff}} = 3.3535$ (against an exact 3.3513, an error of 0.07%) and $Z_0 = 50.642\,\Omega$ (against $50.690\,\Omega$, a difference of $0.048\,\Omega$). The ratio cancels most of the common bias, which is why the engineering quantities are roughly twice as accurate as the capacitance they come from.

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1 Introduction

1.1 What is delivered, and what is not

A microstrip cross-section is specified by geometry and permittivity, and the quantity an engineer wants is the characteristic impedance. Under a lossless quasi-TEM model that impedance follows from a matched pair of capacitances: the dielectric one, C, and the same geometry with every field element assigned air, C_{air}. The deliverable of this project is therefore a number and, with equal weight, the distance from its limit. A single finite-element value that happens to sit near $50\,\Omega$ is not a result; nor is agreement with an empirical formula, nor a claim of convergence inferred from two meshes that happen to agree.

The work is organised around four questions: whether the solver is right (Section 3), what the limit is (Section 4), how far the submitted mesh sits from it and what controls that distance (Sections 5–7), what an adaptive strategy would add (Section 8), and what else contributes at the same size (Sections 9–10). The final quantities follow in Section 11.

1.2 Geometry, materials and notation

The nominal geometry is a copper trace of width $w = 3.0\,\text{mm}$ and thickness $t = 0.035\,\text{mm}$ on a substrate of height $H = 1.6\,\text{mm}$ and relative permittivity $\varepsilon_r = 4.4$, as listed in Table 1 and drawn in Figure 1. All computations use SI units internally. The computational rectangle has half-width W and top-boundary height H_{air}, measured from the ground, so $x \in [-W, W]$ and $y \in [0, H_{\text{air}}]$; the final engineering domain uses $W/H = H_{\text{air}}/H = 8$ after the truncation study of Section 9.

Table 1: Physical and computational parameters for the final microstrip model.

Quantity	Value	Meaning
H	$1.6 \times 10^{-3}\,\text{m}$	substrate height
w	$3.0 \times 10^{-3}\,\text{m}$	trace width
t	$3.5 \times 10^{-5}\,\text{m}$	trace thickness
ε_r	4.4	substrate relative permittivity
V_0	1 V	trace potential
W/H	8	final half-width ratio
H_{air}/H	8	final height ratio

2 Physical and Mathematical Model

2.1 Governing equation and boundary conditions

The field domain Ω is the computational rectangle with the conductor interior removed. The only unknown is the electric potential $\phi(x, y)$ in volts, and the material coefficient $\varepsilon(x, y) = \varepsilon_0 \varepsilon_r(x, y)$ in F/m is piecewise constant: $4.4\varepsilon_0$ in the substrate and ε_0 in air. With no free volume charge,

$$-\nabla \cdot (\varepsilon \nabla \phi) = 0 \qquad \text{in } \Omega, \tag{1}$$

with essential data

$$\phi = V_0 \quad \text{on the signal trace,} \qquad \phi = 0 \quad \text{on the ground plane,}$$

and a homogeneous Neumann condition $\varepsilon \nabla \phi \cdot n = 0$ on the artificial top and side walls Γ_{far}, where n is the outward unit normal. The electric field is $\mathbf{E} = -\nabla \phi$ and the displacement is $\mathbf{D} = \varepsilon \mathbf{E}$; the zero-flux condition therefore prescribes zero normal displacement on the artificial walls, which is an approximation whose cost is measured in Section 10. Across the air–substrate interface the exact conditions are continuity of potential and balance of normal displacement flux. The electrostatic energy per unit length is

$$W_e(\phi) = \frac{1}{2} \int_\Omega \varepsilon(x, y) |\nabla \phi|^2 \, d\Omega, \tag{2}$$

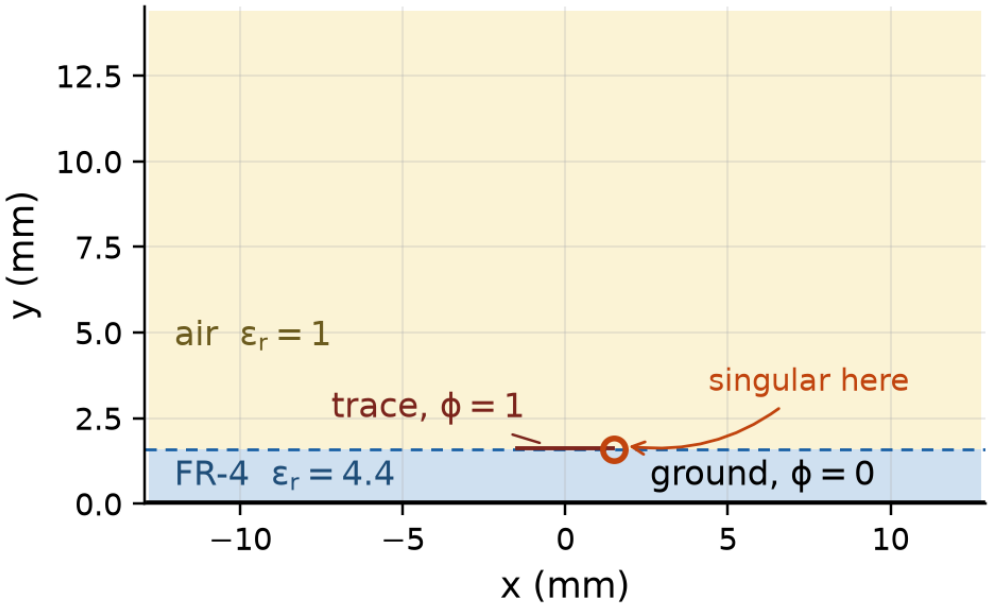


Figure 1: The cross-section. The substrate occupies $0 < y < H$, air lies above it, and the trace interior is removed from the field domain so that its surface is an internal Dirichlet boundary. The circle marks the corner where the conductor edge meets the dielectric interface; it is the only place in the problem where the solution is not smooth, and Sections 6–7 are about it.

and the capacitance follows from it. Positive discontinuous coefficients of this kind are naturally treated in a variational formulation with separate integration over each material region, as discussed by Hackbusch [1, Sec. 10.1].

2.2 Weak form and well-posedness

Let $\partial\Omega$ denote the entire boundary and Γ_D the union of the trace and ground Dirichlet boundaries; $H^1(\Omega)$ contains square-integrable functions with square-integrable first weak derivatives. The affine trial space and homogeneous test space are

$$V_g = \{u \in H^1(\Omega) : u = V_0 \text{ on the trace},\ u = 0 \text{ on the ground}\}, \qquad \mathcal{V} = \{v \in H^1(\Omega) : v = 0 \text{ on } \Gamma_D\}.$$

Multiplying (1) by $v \in \mathcal{V}$ and integrating by parts, with ds denoting boundary arclength, the boundary term vanishes because $v = 0$ on Γ_D and the prescribed normal flux is zero elsewhere. The weak problem is: find $\phi \in V_g$ such that

$$a(\phi, v) := \int_\Omega \varepsilon \nabla\phi \cdot \nabla v \, d\Omega = 0 \qquad \forall v \in \mathcal{V}. \tag{3}$$

The bilinear form is continuous and coercive even across the dielectric jump, since $\varepsilon_0 \le \varepsilon \le 4.4\varepsilon_0$ everywhere:

$$|a(u, v)| \le 4.4\varepsilon_0 \|\nabla u\|_{L^2} \|\nabla v\|_{L^2}, \qquad a(v, v) \ge \varepsilon_0 \|\nabla v\|_{L^2}^2.$$

Because Γ_D has positive boundary measure, the Poincaré inequality holds on $\mathcal{V}$, so the gradient seminorm is a norm there. Taking an H^1 lifting g of the prescribed data and writing $\phi = g + u$ with $u \in \mathcal{V}$ gives a bounded load $\ell(v) = -a(g, v)$, and Lax–Milgram supplies existence and uniqueness. The Dirichlet constraints also remove the constant nullspace of the pure-Neumann problem: two solutions differ by a constant, and a zero Dirichlet trace forces that constant to vanish. The argument is the standard one for elliptic problems, as in Hackbusch [1, Sec. 7.2] and Quarteroni [3, Secs. 3.4 and 4.2].

2.3 Discretisation, assembly and capacitance extraction

The discrete space $\mathcal{V}_h \subset \mathcal{V}$ uses conforming triangles and continuous functions that are affine on each triangle, following Quarteroni [3, Sec. 4.5] and Hiptmair [2, Sec. 2.4]. Writing $\phi_h = \sum_j \Phi_j N_j$ with

nodal basis functions N_j, the stiffness entries are

$$K_{ij} = \int_\Omega \varepsilon \nabla N_i \cdot \nabla N_j \, d\Omega. \tag{4}$$

On a material-conforming triangle of area $|K|$ and coefficient ε_K the basis gradients are constant, so the element contribution $K_{ij}^K = \varepsilon_K |K| \nabla N_i \cdot \nabla N_j$ is integrated exactly; the assembly is therefore exact to roundoff, and the only errors left are discretisation errors. This is the property that makes the verification of Section 3 meaningful: a residual or a matrix identity cannot be "approximately" satisfied.

Prescribed values are imposed by elimination. With Φ_D the prescribed nodal values and Φ_F the unknowns, the reduced system is

$$K_{FF}\Phi_F = b_F, \qquad b_F = -K_{FD}\Phi_D, \tag{5}$$

which is symmetric positive definite. The full reaction vector $r = K\Phi$ carries the surface charge per unit length on the Dirichlet nodes, so the two capacitances are

$$C_E = \frac{2W_e}{V_0^2}, \qquad C_Q = \frac{|Q_{\text{trace}}|}{|V_0|}, \tag{6}$$

where Q_{trace} is the sum of the trace reactions. Agreement of C_E and C_Q tests that energy integration and reaction extraction use the same operator; it is a consistency check, not an accuracy check, a distinction that matters in Appendix A.

Algorithm 1 Microstrip P_1 FEM solver

Require: Material-conforming mesh $\mathcal{T}_h$, geometry, ε_K and voltage V_0

1: Generate the study mesh, or load the exact frozen mesh for a matched replay.
2: Verify the trace hole, interface, element regions and trace/ground/far boundary sets.
3: Set $D = D_{\text{trace}} \cup D_{\text{ground}}$, $F = \{\text{all nodes}\} \setminus D$.
4: Initialise $\Phi_D = V_0$ on the trace and $\Phi_D = 0$ on ground; set $K = 0$.
5: **for** each triangle $K_e \in \mathcal{T}_h$ **do**
6: Compute $|K_e|$ and its three constant basis gradients.
7: Assemble $\varepsilon_{K_e}|K_e|\nabla N_i \cdot \nabla N_j$ into the sparse matrix.
8: **end for**
9: Solve $K_{FF}\Phi_F = -K_{FD}\Phi_D$ by a sparse direct solver.
10: Evaluate $\nabla\phi_h$, W_e, $C_E = 2W_e/V_0^2$ and reactions $r = K\Phi$.
11: Sum trace reactions to obtain Q_{trace} and $C_Q = |Q_{\text{trace}}|/|V_0|$.
12: Check bounds, symmetry, residual, charge balance and C_E/C_Q consistency.

Ensure: Potential, capacitances and diagnostics; failed checks disqualify interpretation.

The recorded environment uses Python 3.13 with scikit-fem 12.0.2 for mesh topology, quadrature and the manufactured-solution basis, and SciPy sparse direct solves. Our own code defines the trace hole, the material and boundary tags, the grading and the capacitance extraction; those parts are tested separately from the library, in `code_hardened/`, so that a library change cannot silently move the answer.

3 Solver Verification

3.1 Twenty-three analytic configurations

Before any mesh study, the operator and the extraction are checked against cases whose answer is known in closed form. Figure 2 groups the configurations into five families: a uniform field against $\phi = y/d$ for $\varepsilon_r = 1$ and 4.4; a parallel-plate capacitance against $\varepsilon_0\varepsilon_r A/d$; the two-dimensional series-layer capacitance; continuity of the normal displacement across an interface; and the total induced

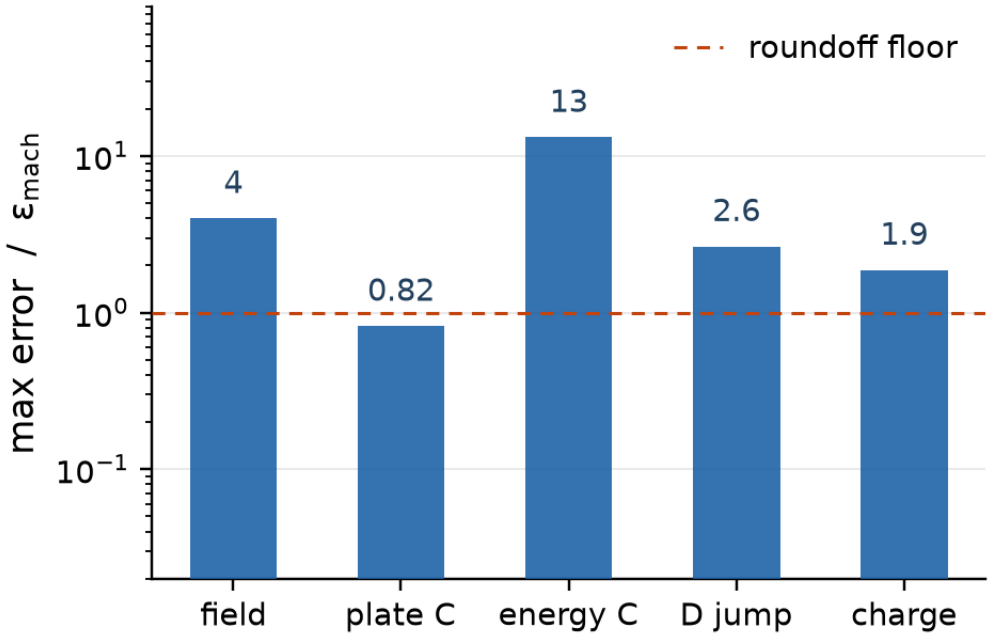


Figure 2: Worst error in each family of analytic checks, in units of machine epsilon (23 configurations in total). The dashed line is the roundoff floor. Every family sits within a factor of about thirteen of it, so the discrete operator reproduces the exact identities of the model rather than approximating them.

charge implied by charge balance. Each configuration is run through the production assembly and the production extraction, not through a test-only copy.

All 23 checks pass, the worst relative error being about $13\varepsilon_{\mathrm{mach}}$. Two consequences are used later. No part of the reported error can be attributed to assembly, extraction or the Dirichlet treatment in the constant-coefficient case, so what remains is discretisation and truncation. And a check that fails at this level is a coding error rather than a modelling choice, which is why the guard stays in the pipeline instead of being run once.

3.2 Manufactured solution

A manufactured solution measures the discretisation error directly and compares the observed rate with the smooth-solution estimate. On the unit square with zero Dirichlet data we impose $u(x,y) = \sin(\pi x)\sin(\pi y)$ and $f = 2\pi^2 \sin(\pi x)\sin(\pi y)$ in $-\Delta u = f$, and refine five times. Table 2 reports the L^2 and energy errors together with the pairwise orders $p_i = \log(E_{i-1}/E_i)/\log(h_{i-1}/h_i)$.

Table 2: Consecutive manufactured-solution refinements. N_T is the triangle count and r_F the free-system algebraic residual; all levels use a sparse direct solve.

Level	N_T	DOFs	$h_{\max}$	$\|e\|_{L^2}$	$\|e\|_E$	$\|r_F\|_\infty$	p_{L^2}	p_E
2 (h)	32	25	0.35355	7.596×10^{-2}	8.397×10^{-1}	3.331×10^{-16}	–	–
3 ($h/2$)	128	81	0.17678	2.041×10^{-2}	4.320×10^{-1}	5.551×10^{-16}	1.896	0.959
4 ($h/4$)	512	289	0.08839	5.201×10^{-3}	2.176×10^{-1}	9.021×10^{-16}	1.972	0.989
5 ($h/8$)	2048	1089	0.04419	1.307×10^{-3}	1.090×10^{-1}	1.457×10^{-15}	1.993	0.997
6 ($h/16$)	8192	4225	0.02210	3.270×10^{-4}	5.451×10^{-2}	2.368×10^{-15}	1.998	0.999

Least-squares fits over the finest three levels give $p_{L^2} = 1.9956$ and $p_E = 0.9983$, and the largest residual listed is 2.4×10^{-15}, far below the acceptance level $\|r_F\|_\infty < 10^{-10}$ used throughout the project. The protocol is the one of Hiptmair [2, Sec. 3.8], and the rates are the ones Quarteroni predicts for these elements [3, Sec. 4.5]. What the table supports is the implementation that was tested, not the microstrip geometry; the geometry tags carry their own test, and the corner behaviour

of Section 6 is measured rather than assumed.

3.3 Layered interface

The layered problem checks the treatment of a discontinuous coefficient, which is the one feature the manufactured solution does not exercise. On the unit square, with $a = 1$ for $x < 1/2$ and $a = 10$ for $x > 1/2$ and $f = 0$, the exact solution is

$$u_*(x) = \begin{cases} 20x/11, & x \le 1/2, \\ (2x + 9)/11, & x \ge 1/2, \end{cases}$$

for which both traces equal $10/11$ at the interface while the gradients are $20/11$ and $2/11$, so the physical flux $q_* = -a\partial_x u_* = -20/11$ is continuous. The test calls the production `assemble_stiffness` with relative coefficients 1 and 10 and divides out the common ε_0.

Table 3: Layered verification using the production assembly. Odd N leaves the interface unresolved and tags elements by centroid: it is a negative control, not an acceptable method.

N	Fitted?	DOFs	$\|u_h - u_*\|_\infty$ (nodes)	$\|q_h - q_*\|_{L^2}$
4	yes	25	3.33e-16	3.18e-15
8	yes	81	8.88e-16	1.04e-14
16	yes	289	2.11e-15	3.86e-14
32	yes	1089	3.22e-15	9.15e-14
7	no	64	8.15e-02	1.07e+00
15	no	256	3.95e-02	5.80e-01
31	no	1024	1.94e-02	3.42e-01

On fitted meshes the largest normal-flux jump is 2.28×10^{-13}, and the reduced matrices are symmetric with a positive smallest eigenvalue. The negative control carries the lesson. It holds a small residual and a symmetric positive matrix while its solution error is ten orders of magnitude larger, so solving the discrete equations accurately says nothing about whether the interface was represented. That is the failure mode a coefficient-jump bug produces.

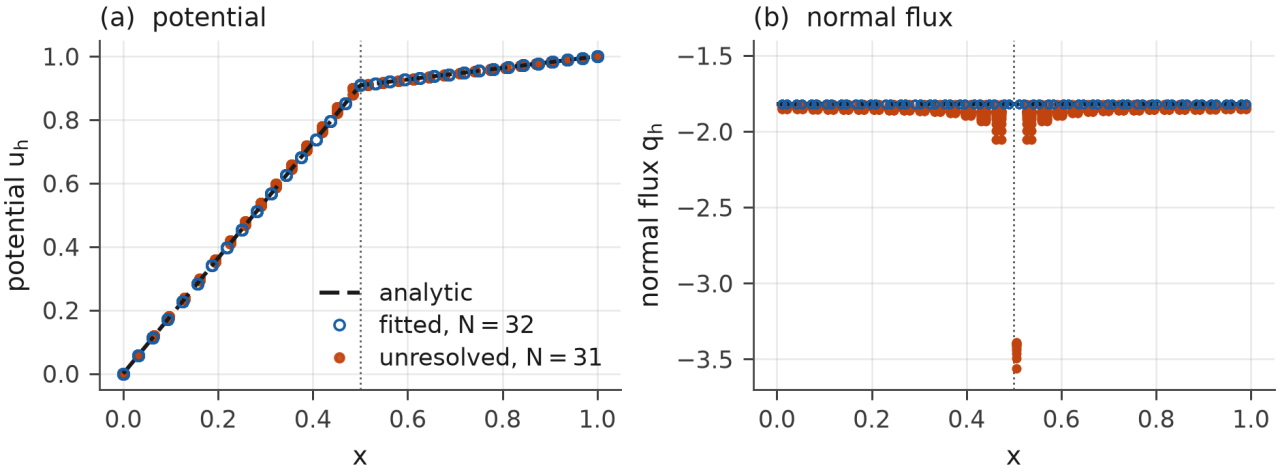


Figure 3: Layered interface with $a = 1$ for $x < 1/2$ and $a = 10$ for $x > 1/2$. The fitted mesh ($N = 32$) reproduces the analytic potential and keeps the normal flux $q_h = -a\,\partial_x u_h$ at its exact value $-20/11$ on both sides of the interface. The unresolved control ($N = 31$) tracks the potential almost as well and yet its flux jumps to -3.4 at the interface, which is the paired failure the table above records in numbers.

3.4 The Dirichlet row on non-zero data

The checks above all use boundary data with the ground at 0 V, and so does the production API, which hard-wires that value. On such data the ground columns of the elimination shift $-K_{FD}\Phi_D$ are identically zero, so no test in the suite can see an error confined to them. We therefore verified the row separately (`code_hardened/boundary_row_audit.py`, commit `ccd2f37`) by building the same system three independent ways: the shipped elimination, a textbook row replacement, and the shipped solver shifted by a constant, on a layered box driven with 1 V on the ground and 3 V on the trace.

The three constructions agree to 8.1×10^{-16} of the applied voltage on the box and to 7.8×10^{-15} on the project mesh, the row-replacement solution matches the closed-form series field to 1.1×10^{-15}, and one element is eliminated by hand in exact rational arithmetic with error zero. Deleting the ground columns of the shift, which is the error the observation is about, moves the same norm by 0.33 V on the box. On the project's own mesh it moves the norm by 7.8×10^{-15} V, which states the blind spot quantitatively: it is real, and the ordinary suite cannot reach it.

4 The Limit

4.1 Two independent routes

Every error quoted in this report is an error against a limit, so the limit has to be fixed before anything is measured. We use two routes that share neither element type nor code. The first is the project's own P_1 sequence on the nested $8H$ meshes M3–M7, extrapolated with the Richardson formula for a halving sequence,

$$C_{\text{ext}} = C_k + \frac{C_k - C_{k-1}}{2^{p_C} - 1}, \qquad p_C = \log_2 \frac{C_{k-2} - C_{k-1}}{C_{k-1} - C_k}, \tag{7}$$

which uses only the fact that successive differences shrink geometrically. The second is a separately written P_2 solver built on scikit-fem's element and assembly, run on the same mesh family, so that a shared bug in our own assembly would move only one of the two limits.

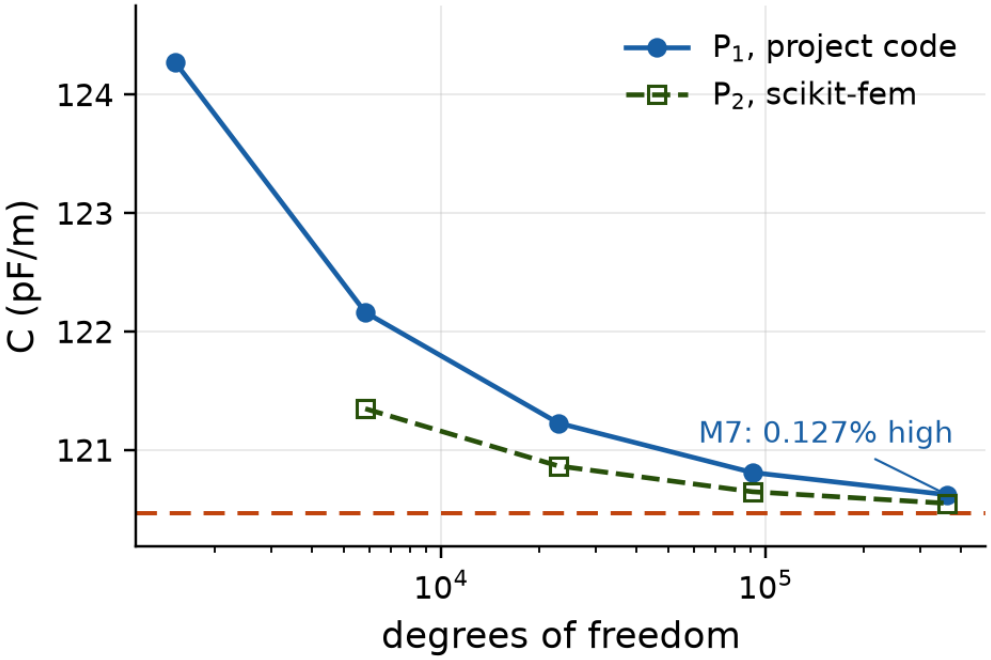


Figure 4: Both families against the adopted limit. The two curves have different element types, different unknowns and different codes, and both approach the same dashed line from above, the P_2 sequence sitting closer to it at equal unknowns.

The two extrapolations are 120.4673 pF/m (P_1) and 120.4670 pF/m (P_2). They agree to 2.9×10^{-6} relative, which is three orders of magnitude below the effect we are about to measure, so the adopted reference

$$C_{\text{ref}} = 120.467 \text{ pF/m} \tag{8}$$

is not a fit to either family but the midpoint of two independent limits. This is the only place where the two solvers are combined; every later number is produced by the P_1 code alone.

4.2 Why a single family would not be enough

A single extrapolation cannot distinguish a limit from a slowly drifting artefact: if the observed order is not constant, formula (7) reports a number whose plausibility rests entirely on the assumption it uses. The second family supplies an independent test of that assumption rather than more digits. What counts is that the two answers agree although their orders differ slightly, 1.16 against 1.15 (Section 6): both families are limited by the same non-smooth feature, so both extrapolations head for the same limit, yet neither has reached the asymptotic regime in which the nominal order would apply.

5 What the Old Metric Measured

5.1 Change is not error

The convergence indicator used earlier in the project was the relative change between consecutive meshes, $dC_k = |C_k - C_{k-1}|/|C_k|$. At M7 it is 0.156%, below the ±0.1% band the specification suggests, and that was read as convergence. Measured against the reference of Section 4, the distance is 0.127%, more than twice as large.

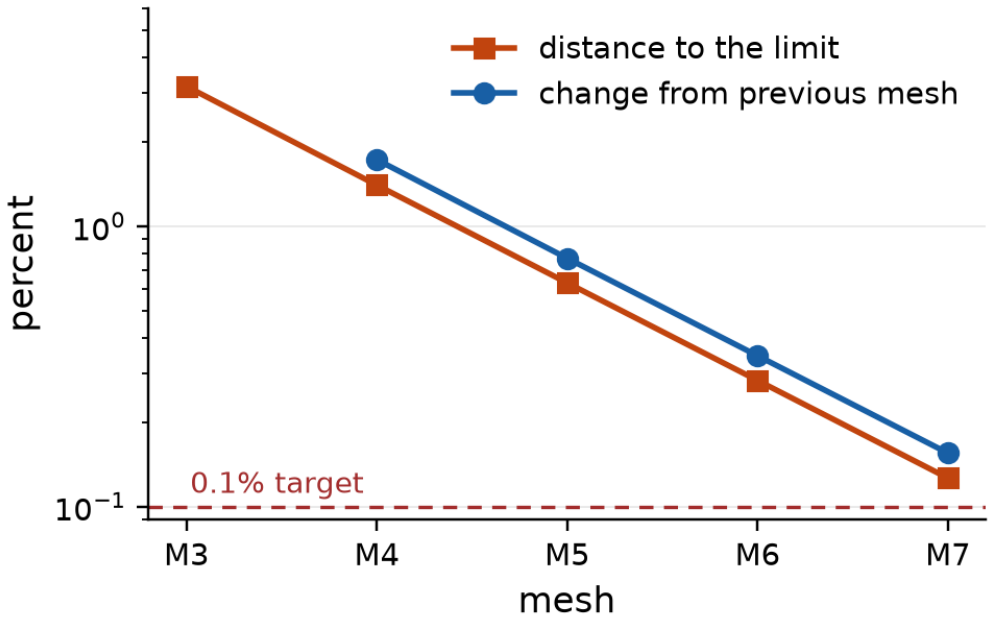


Figure 5: The two quantities the report has to keep apart. The change from the previous mesh falls below the 0.1% target at M6–M7 while the distance to the limit is still about 0.13%. A declining difference says the sequence is settling; it does not say how far from its limit it is.

The relation between the two is fixed by the order. For a sequence with error $e_k \propto h_k^p$ and mesh halving,

$$dC_k \simeq (2^p - 1)\, e_k,$$

so the change is the error multiplied by $2^p - 1$. With $p = 1.16$ that factor is 1.23, which reproduces 0.156/0.127 to the quoted digits. The factor is below one here, so the change *understates* the error;

for $p > 1$ it would overstate it. Without an estimate of p, a shrinking difference therefore bounds nothing.

Table 4: The uniform sequence against the reference of (8). The last two columns are the two quantities of Figure 5: the change from the previous mesh, and the actual distance from the limit.

Mesh	DOFs	C_E (pF/m)	change (%)	error (%)	dC/error
M3	1,512	124.273	–	3.16	–
M4	5,840	122.159	1.731	1.40	1.24
M5	22,944	121.225	0.770	0.63	1.22
M6	90,944	120.807	0.346	0.282	1.23
M7	362,112	120.620	0.156	0.127	1.23

The ratio in the last column is constant at 1.23, as it must be while the sequence stays in its geometric regime. That constancy is what makes the 0.127% a measured distance rather than an artefact of one particular mesh pair, since the same factor relates every consecutive pair in the table.

5.2 Consequence for the submitted value

M7 misses the ±0.1% band by a factor of about 1.27, and the honest statement is that the submitted capacitance is 0.127% from its limit rather than converged to 0.1%. Two responses are open: refine further, or change the discretisation so that the same number of unknowns buys a smaller error. M8 would be roughly 1.4 million unknowns behind a sparse direct factorisation, so Section 7 takes the second route.

6 Why the Order Is 1.16

6.1 Quadratic elements do not buy an order

If the slow convergence were a property of the basis, doubling the polynomial degree would remove it. Figure 6 fits the error in C against an h proxy for both families: the slope is 1.16 for P_1 over 1.5k–362k unknowns and 1.15 for P_2 over 5.8k–362k. The two are the same to the quoted digits.

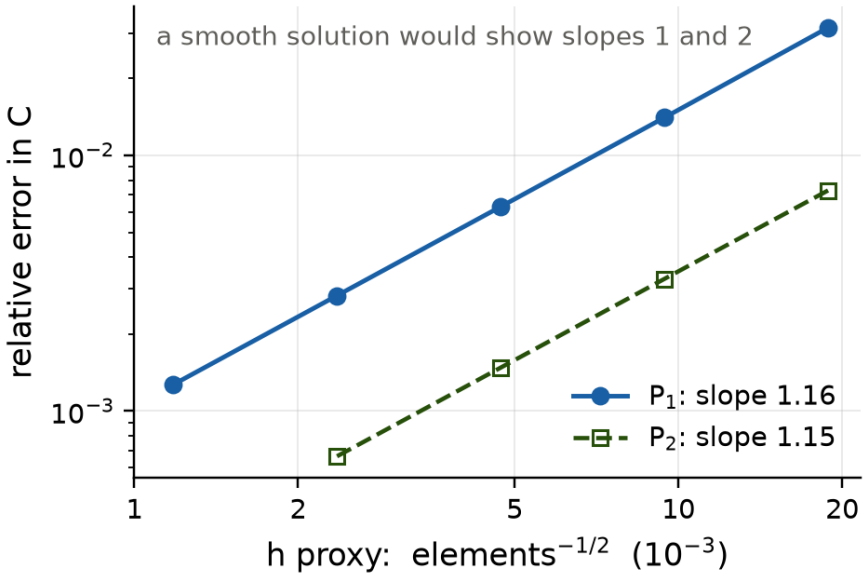


Figure 6: Error in C against an h proxy for both families. A smooth solution would give slopes of one and two for P_1 and P_2 respectively; both families instead lie on nearly the same line. The offset between them is a constant, not a rate.

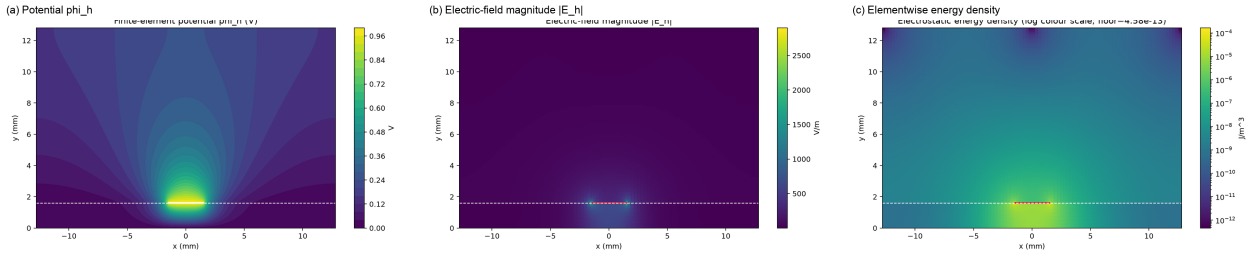


Figure 7: Baseline field diagnostics on the production $8H$ domain: (a) finite-element potential ϕ_h, (b) electric-field magnitude $|\mathbf{E}_h|$ with $\mathbf{E}_h = -\nabla\phi_h$, and (c) elementwise electrostatic energy density $\frac{1}{2}\varepsilon|\mathbf{E}_h|^2$ (J/m^3), shown with a logarithmic colour scale for visibility. The potential respects the imposed Dirichlet values, while the field and the energy density concentrate near the trace corners; away from the trace the box is quiescent. These plots are diagnostic visualisations, not error norms.

The diagnosis is unambiguous. A P_2 approximation has twice the local polynomial degree and four times the unknowns per element, yet it converges at the same rate, so what limits the error is the regularity of the solution and not the approximation space. Spending unknowns on a richer basis is therefore not a cure; the error is generated at a point. Figure 7 shows the point. On the production domain the potential is flat over most of the box and the field is confined to the dielectric within a few hundred micrometres of the trace edge, so the unknowns that sit elsewhere have nothing to resolve.

6.2 The corner sets the rate

The point is the corner marked in Figure 1, where the conductor edge terminates on the dielectric interface. There the local solution behaves like r^α with $\alpha < 1$, so the gradient is unbounded and the usual h^p-type estimate does not apply. Fitting ϕ_h along rays from each of the four trace corners on the two finest meshes gives a median exponent $\alpha = 0.52$ (range 0.47–0.61 across corners and levels). The exponent has a field consequence that does not depend on the fit: if the potential behaves like r^α then $|\nabla\phi|$ behaves like $r^{\alpha-1}$, so the peak value keeps moving under refinement instead of settling. That is visible in Figure 7(b), where the saturated core is one or two elements wide at every mesh size used here, and it is why no local field strength is reported as a design quantity: the number belongs to the discretisation, not to the device.

For a quadratic functional of the solution the expected rate is 2α, so 1.03 is the prediction to set against the measured 1.16. The match is of the right order rather than exact, and it cannot be better: the constant multiplying the singular term, the mixture of four corners with different exponents and the dielectric interface all enter the prefactor, and the fit window is not asymptotic. What the comparison does establish is the mechanism, since the measured rate is neither one nor two but twice a corner exponent strictly between 0 and 1. The same value holds for both element families, because both resolve the same singularity.

7 Corner Grading

7.1 What is refined, and what is not

If the error is generated at the corners, the mesh should be fine there and coarse elsewhere. Figure 8 makes that visible on a pair of meshes small enough to see: the uniform M2 mesh and the graded G2 mesh in the same window, then the same pair enlarged at the right-hand trace end. G2 costs 1,422 unknowns against 404 for M2, and every extra element sits within a few hundred micrometres of the four corners. The middle of the trace and the far field are the same elements in both pictures. That is the whole claim of grading, separated from any question of cost: it moves resolution, it does not add physics.

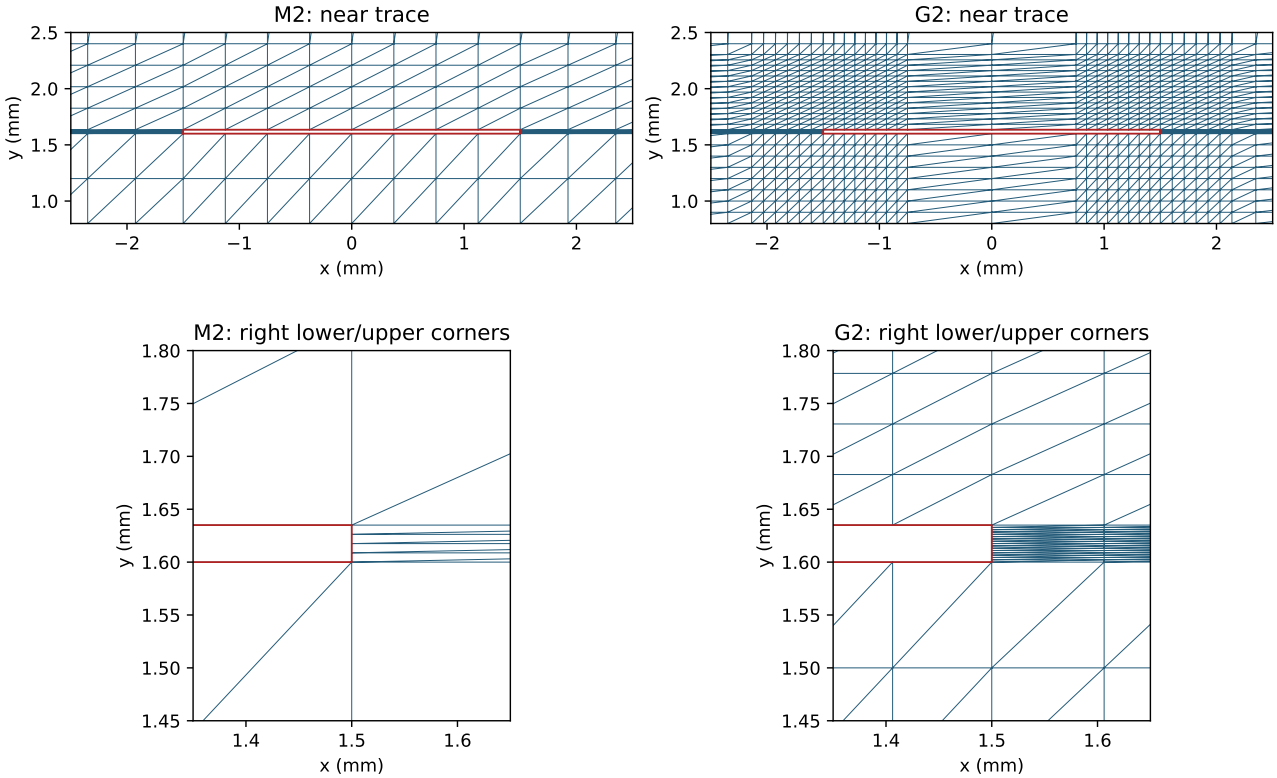


Figure 8: What grading changes and what it leaves alone. Top row: uniform M2 and graded G2 in the same window near the right-hand trace end. Bottom row: the same pair enlarged at the corner. The two meshes differ only within a few hundred micrometres of the corner: G2 has 1,422 unknowns against 404 for M2, and every extra element is spent there. These are geometry views and not a matched-DOF efficiency claim.

The same rule drives the production run. On the fixed M5 background, round k marks the elements within $r_0 0.35^{k-1}$ of a corner, with $r_0 = 1.6 \times 10^{-4}$ m, and the marked elements are refined by newest-vertex bisection with a symmetry completion that keeps the mesh conforming about $x = 0$. Nothing else changes: the operator, the boundary conditions, the extraction and the material tags are those of Algorithm 1. Seven rounds of this produce the mesh behind the numbers in Section 7.3, and at the seventh round the corner elements are about $0.3\,\mu$m across, which is under one printed pixel at the width of a page. That is why Figure 8 shows the two-round mesh: the last rounds are visible as a number, not as a picture.

7.2 The plateau moves with the background

The graded error does not fall to zero on a fixed background. It stalls at a plateau set by the resolution far from the corner, and repeating the same seven rounds on four backgrounds separates the two contributions: the plateau falls by a factor of about four per background level (Table 5).

Table 5: Final error after the same grading campaign on four backgrounds. The plateau ratio is the previous row divided by the current one; a value near four is the signature of an h^2 error in the far field.

Background	final DOFs	error (%)	plateau ratio	DOFs ratio
M2	2,560	1.087	–	–
M3	9,772	0.270	4.02	3.82
M4	36,396	0.069	3.91	3.72
M5	139,426	0.017	4.12	3.83

The plateau ratios (4.02, 3.91, 4.12) are constant, so the residual error is a second-order error of the background mesh and not a numerical floor. Because the unknowns grow by the same factor as the plateau falls, one further order of magnitude costs a full background refinement: grading buys a constant factor, not a better rate.

7.3 What the grading is worth

On the M5 background the campaign ends at 139,426 unknowns with 0.017%, against 362,112 unknowns and 0.127% for uniform M7. That is 7.6× closer at 39% of the unknowns, and it is the reason the graded value is reported as an alternative discretisation of the same model. It is not substituted for the M7 value: the submitted capacitance is the uniform one, because that is the sequence against which the reference in Section 4 was built, and mixing the two would destroy the error attribution the rest of the report depends on. The campaign behind these numbers, including what happens on the other three backgrounds, is Figure 9.

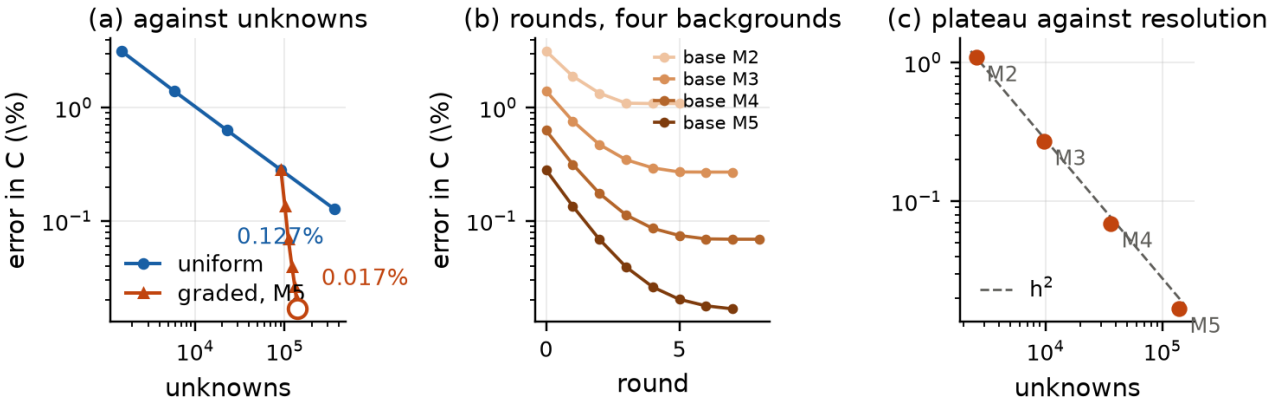


Figure 9: The campaign on three axes. (a) Error in C against unknowns for both families: the graded sequence ends at 139,426 unknowns and 0.017%, the uniform M7 point at 362,112 unknowns and 0.127%. (b) The same seven rounds repeated on four backgrounds, which stall at a plateau instead of reaching zero. (c) That plateau against the resolution of its own background, with the h^2 reference line: the plateau follows the background, not the number of rounds.

8 Residual-Based Adaptive Refinement

8.1 A second error measure, declared first

Grading is a rule we imposed. The alternative is to let the computed solution choose the elements, so this section runs a residual-based adaptive campaign on the same problem and compares it with uniform and graded refinement. One thing must be said before any number is quoted: this section measures a different quantity from Sections 4 and 7. The capacitance experiment asks how far the computed C is from the Richardson limit, in per cent; the experiment here asks how far the computed potential is from a frozen reference solution, in the energy norm. A small energy error does not imply a small capacitance error, and the meshes, the starting points and the stopping rules all differ. Every number below is an energy-norm number, and nothing here changes the submitted capacitance.

Table 6: The two error measures of this report, kept apart: they answer different questions and are not interchangeable.

	Energy norm (Section 8)	Capacitance (Sections 4 and 7)
Error	$E_{\mathrm{rel},M6} = \|e_{M6}\|_E / \|\phi_{M6}\|_E$	$\|C_h - C_{\mathrm{ref}}\| / C_{\mathrm{ref}}$
Reference	frozen M6 and M7 solutions	Richardson limit of the $12H$ family
Setting	M1-S start, $\theta = 0.5$, eight cycles	M5 background, seven rounds; uniform M2–M7

8.2 Where the indicator puts the elements

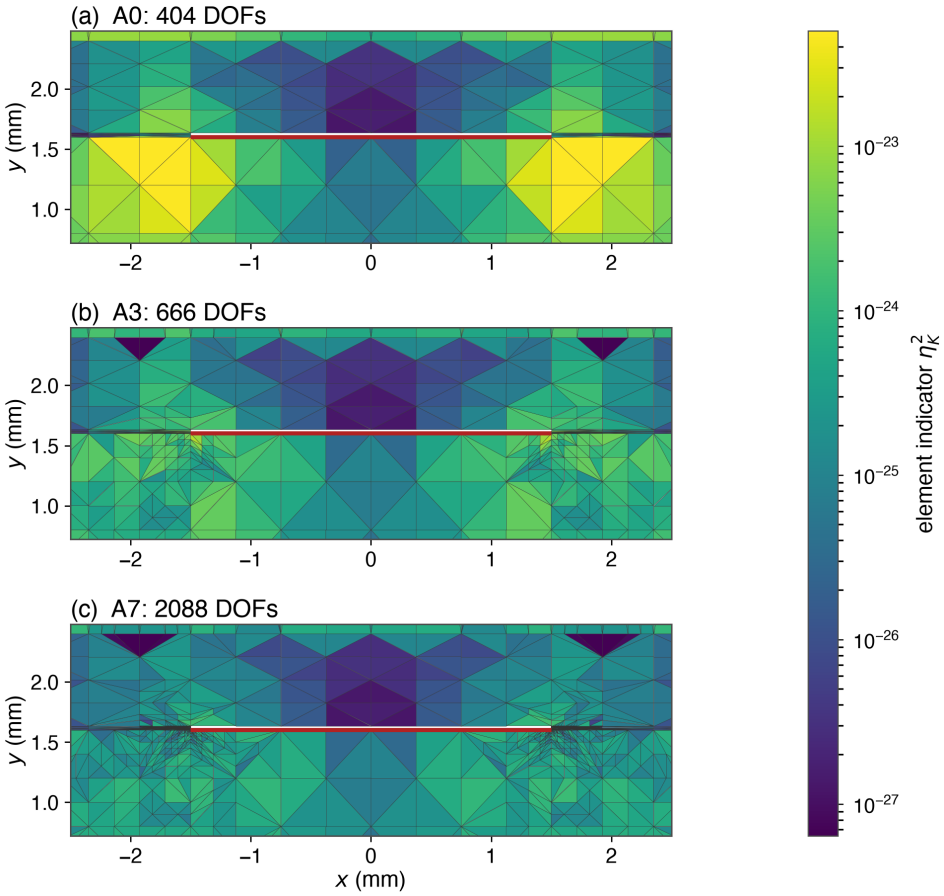


Figure 10: Where the indicator puts the resolution: the A0, A3 and A7 meshes in the same physical window near the right-hand trace end, with the conductor edge in red. Element colours show η_K^2 on one shared logarithmic scale, so the three panels are directly comparable. Refinement collects at the trace ends, but not only there: marked elements also appear in the substrate and in the air above the corner. This is observed localisation, not a claim that the indicator identifies every singular contribution.

For $f = 0$, constant elementwise ϵ and affine $\phi_h|_K$ the interior volume residual vanishes, so the indicator is built entirely from the jumps of the flux across interior edges,

$$\eta_K^2 = \frac{1}{2} \sum_{e \subset \partial K \cap \mathcal{E}_{\text{int}}} h_e \|J_e\|^2_{L^2(e)}, \qquad J_e = (\epsilon \nabla \phi_h|_{K^+} - \epsilon \nabla \phi_h|_{K^-}) \cdot n_e,$$

with one normal n_e used on both sides and interface edges included. This is a residual indicator, not a guaranteed error bound. At every cycle the elements are sorted by η_K^2 and the smallest prefix carrying at least $\theta = 0.5$ of $\sum_K \eta_K^2$ is refined; reflected elements are added to keep the mesh symmetric about $x = 0$, and every split edge is closed so that the triangulation stays conforming. Eight cycles are run from a symmetry-completed M1 start with 404 unknowns.

Within $0.25H$ of a trace corner the element fraction rises from 5.11% at A0 to 62.0% at A7, and those elements carry 79.6% of the final squared indicator; the far boundary carries 0.00265% of it, so no resolution is being wasted on the outside box. Two features of Figure 10 are worth naming. The refinement is not confined to the four corners we would have picked by hand: marked centroids reach $|x| = 5.0\,\text{mm}$ and $y = 4.57\,\text{mm}$, outside the corner disks, because the substrate and the air above the trace end carry a real flux imbalance too. And the concentration is strongest early. At A0, 88.6% of the indicator already sits in elements whose nearest corner is a lower junction; by A7 that fraction has fallen to 50.2%, so lower-junction dominance is an early-cycle effect rather than a permanent property of the indicator.

8.3 What it buys, and what it costs

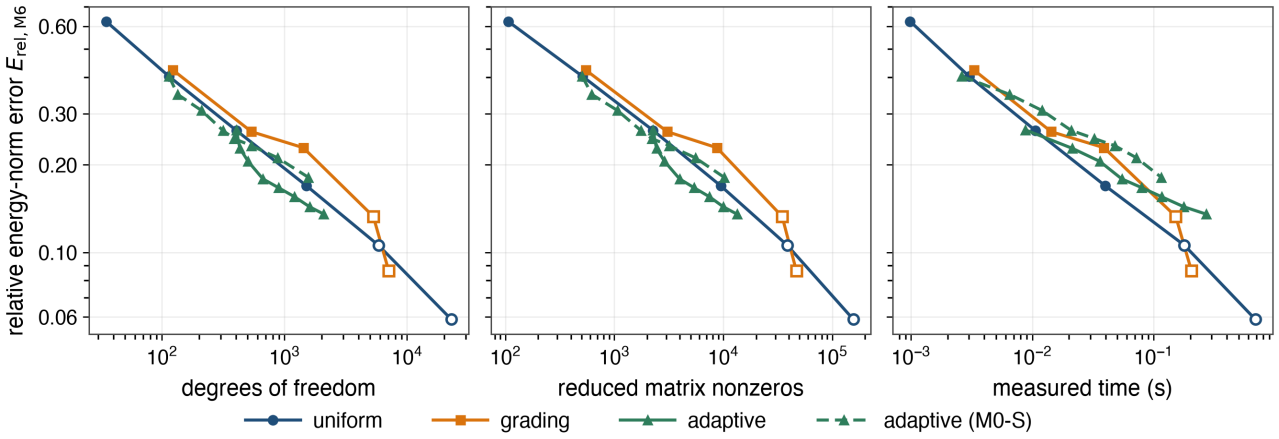


Figure 11: All three refinement strategies in one error measure, $E_{\mathrm{rel,M6}}$, against (a) unknowns, (b) reduced matrix nonzeros and (c) measured median total algorithm time. Adaptive work is cumulative from its initial mesh, so each point contains every preceding solve, estimate, mark and refine cycle. Open markers fail the 3% M6/M7 reference-sensitivity test and are shown for completeness, not for formal comparison.

Panel (a) is the flattering one for adaptivity. At a common target of $E_{\mathrm{rel}} = 0.30$ the adaptive path needs about 224 unknowns against 273 for uniform refinement and 349 for grading, so the DOF ratio is 1.218 and the nonzero ratio 1.235 in its favour. Panel (c) is the other half of the same experiment: at that target the cumulative median time is 13.1 ms for the adaptive path against 7.15 ms for uniform and 9.30 ms for grading. Adaptivity wins on unknowns and loses on time, by a factor of about 1.8 against uniform.

The direct comparison, which needs no interpolation, says the same. A5 reaches $E_{\mathrm{rel}} = 0.1553$ with 1,206 unknowns while uniform M3 reaches 0.1694 with 1,512: adaptive is 20% smaller and takes 2.9 times the cumulative time. Nor is the advantage in unknowns uniform along the curve. At $E_{\mathrm{rel}} = 0.40$ the DOF ratio is 1.013, which is no saving at all, and at the fine end A7 needs 2,088 unknowns where uniform M3 already had 1,512 at a comparable error.

The mechanism is in how the work is counted, and it is not a bookkeeping convention. A uniform mesh is one mesh: generate, assemble, solve once. An adaptive path is eight solves, seven estimator evaluations, seven marking passes and seven refinements, and the cumulative time at cycle k contains all of them, because all of them are needed to obtain the mesh that produced the error at cycle k. The saving in memory is real, since the final adaptive mesh is smaller than the uniform mesh of the same error. The cost in time is equally real, because the coarse solves cannot be skipped to reach the fine one. At millisecond scale none of this separates cleanly from machine noise, so the claim is kept qualitative: residual adaptivity reduces unknowns on this problem, and it does not reduce wall time.

For the report the practical consequence is that grading remains the working method. It is a rule that fits in one line, it needs no reference solution to run, and it costs one mesh generation and one solve. The adaptive campaign is kept for a different reason: it locates the error without using our own guess about where the error is, and it agrees with that guess.

9 Algebraic Error

9.1 A ladder of tolerances

Every number so far is a discretisation error, which assumes the linear systems are solved exactly enough not to matter. That assumption is tested rather than asserted. On M5 the sparse direct solve is replaced by conjugate gradients with relative tolerances from 10^{-2} to 10^{-12}, and the resulting change in C is compared with the discretisation error on the same mesh, 6.3×10^{-3}.

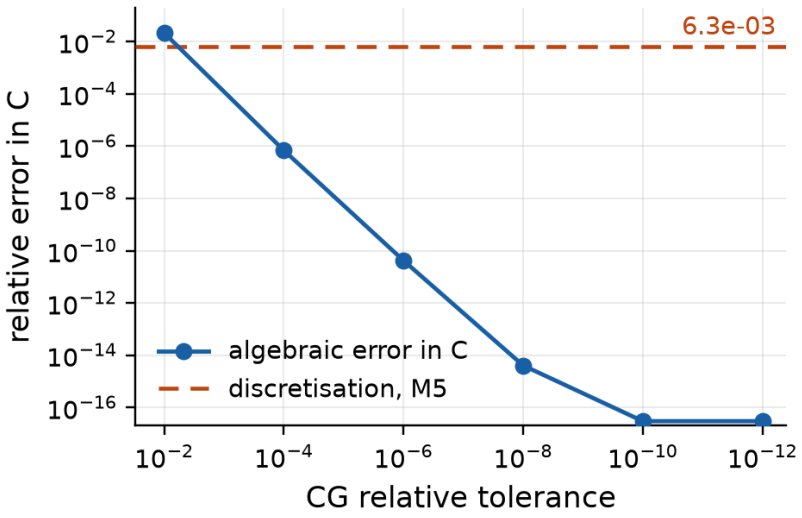


Figure 12: Algebraic error in C against the CG stopping tolerance on M5. The error follows the tolerance until about 10^{-8} and then stops, at which point the discretisation error of the mesh is the only thing left.

Table 7: Stopping tolerance against the induced error in C on M5. The last column compares it with the discretisation error of the same mesh.

CG rtol	error in C	scaled residual	error / 6.3×10^{-3}
10^{-2}	2.1×10^{-2}	9.4×10^{-3}	3.4
10^{-4}	6.9×10^{-7}	5.2×10^{-5}	1.1×10^{-4}
10^{-6}	4.2×10^{-11}	1.1×10^{-6}	6.7×10^{-9}
10^{-8}	4.1×10^{-15}	5.7×10^{-9}	6.4×10^{-13}
10^{-10}	0	6.2×10^{-11}	0

Once the tolerance is below 10^{-8} the two errors are separated by about thirteen orders of magnitude: the algebraic contribution sits at the roundoff floor, 1×10^{-16}, while the discretisation error is 6.3×10^{-3}. Tolerance is not irrelevant, since at 10^{-2} the algebraic error is three times the discretisation error and would corrupt every number in Section 5, but at the default setting it is a solved problem and the remaining error budget belongs to the corner.

10 Domain Truncation

The unbounded air region is approximated by a finite rectangle with a zero-flux outer boundary. To isolate the effect we used the sequence $(W/H, H_{\text{air}}/H) = (3,3), (5,5), (8,8), (12,12)$ while preserving the near-trace mesh, and repeated the largest domain with a finer near-field resolution to separate truncation from mesh resolution.

Table 8: Domain-truncation sequence. The last row repeats $12H$ with a finer near field to show that the $12H$ value is itself resolved to better than 0.04%.

$W/H = H_{\text{air}}/H$	Elements	DOFs	C_E (pF/m)	difference from $12H$
3	11868	6121	118.839	2.49%
5	16988	8711	121.004	0.71%
8	26168	13346	121.653	0.18%
12	41208	20926	121.875	reference
12 (refined)	60720	30741	121.831	0.036%

The $8H$ box costs 0.18%, the same size as the discretisation error at M7 (0.127%).

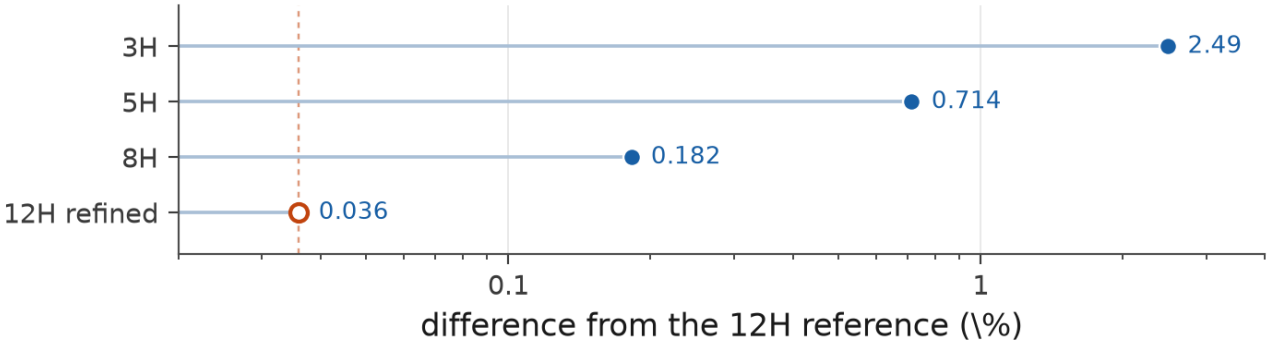


Figure 13: Difference from the $12H$ reference for each domain. The open marker and the dashed line mark the refined $12H$ repeat, whose 0.036% is the mesh-noise floor. On a logarithmic scale the decay is close to exponential in the box size before it flattens at that floor.

Two caveats qualify that statement rather than sit in a footnote. The $8H$ and $12H$ values in Table 8 are production meshes at their own resolutions, so their difference mixes truncation with mesh resolution; the refined $12H$ repeat puts the latter at 0.036%, which is why the 0.18% is attributed mainly to truncation. And the M7 value of Section 11 is not on this sequence at all, so 0.18% measures the sensitivity of the answer to the domain choice; it is not a measured truncation error for the submitted value. It is quoted beside the discretisation error to show that the two are comparable, not to be added to it.

11 Final Values

11.1 Capacitance and its distance from the limit

The submitted dielectric capacitance is the M7 value,

$$C_{M7} = 120.620 \text{ pF/m}, \tag{9}$$

and its distance from the extrapolated limit $C_{\text{ref}} = 120.467\,\text{pF/m}$ of (8) is

$$C_{M7} - C_{\text{ref}} = 0.153 \text{ pF/m}, \tag{10}$$

which is 0.127% of C_{ref}. Both numbers are written out rather than folded into a $\pm$ interval, because 0.153 is not an uncertainty: the M7 solve is deterministic and repeatable, and the gap is what the mesh resolution still costs against the limit. The finite box adds a separate sensitivity of 0.18%, discussed in Section 10, which is of the same size; the two are reported side by side and are never added into one error bar. The all-air run reuses the exact frozen M7 mesh — same nodes, same boundary sets, same sparsity pattern, only the material tags changed — giving $C_{\text{air}} = 35.968\,\text{pF/m}$, which is 0.06% above its own extrapolated limit of $35.946\,\text{pF/m}$.

Reusing the mesh is not a convenience. If the all-air case were remeshed, the ratio C/C_{air} would contain a mesh difference as well as a material difference, and the effective permittivity would inherit an error that no convergence study of C alone could detect. The mesh fingerprint is checked before the air run, so the material change is the only difference between the two systems.

11.2 Effective permittivity

The ratio

$$\varepsilon_{\text{eff}} = \frac{C}{C_{\text{air}}} = 3.3535 \tag{11}$$

is to be compared with the exact ratio of the two limits, 3.3513. The discrepancy is 0.07%, about half the 0.127% error of C itself, because the two errors are strongly correlated: both meshes share the same corner under-resolution, and dividing removes the common part. The improvement is real, but it is not independent information. The same bias is cancelling, not being measured twice.

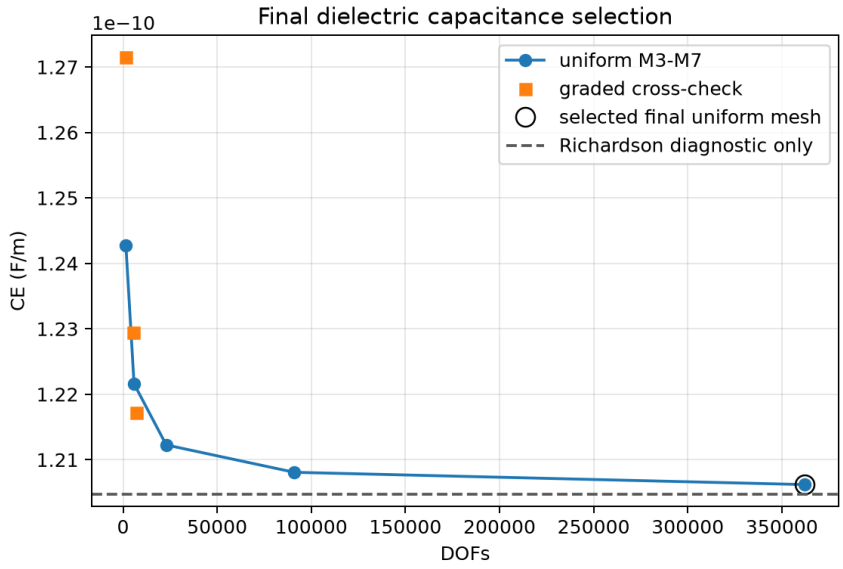


Figure 14: Final dielectric capacitance selection. M7 is the selected computed FEM mesh; the graded run is an independent cross-check on the same quantity, and the Richardson line is a diagnostic only.

11.3 Characteristic impedance

With c_0 the vacuum speed of light, the quasi-TEM relations give

$$L = \frac{1}{c_0^2 C_{\text{air}}}, \qquad Z_0 = \sqrt{\frac{L}{C}} = \frac{1}{c_0\sqrt{C\,C_{\text{air}}}} = \frac{1}{c_0 C_{\text{air}}\sqrt{\varepsilon_{\text{eff}}}}. \tag{12}$$

The three expressions agree to floating-point precision, which is an internal identity and not a validation. Numerically, $Z_0 = 50.642\,\Omega$ against $50.690\,\Omega$ from the reference pair, a difference of $0.048\,\Omega$ or 0.09%.

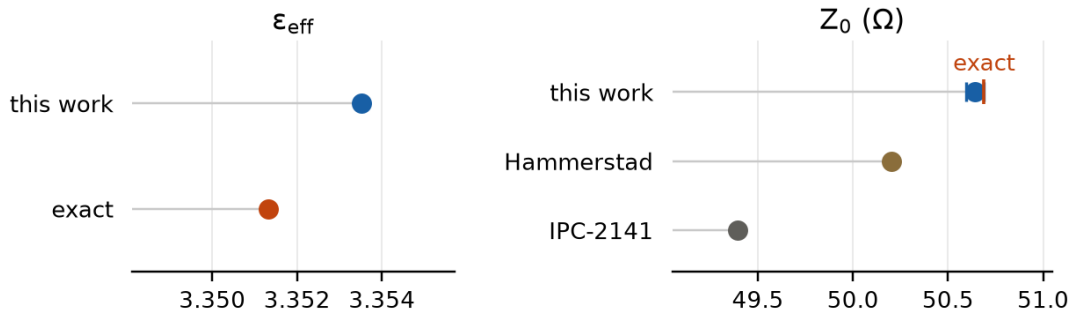


Figure 15: The two engineering quantities against their exact counterparts and the independent formulas. Left: ε_{eff} against the ratio of the extrapolated limits. Right: Z_0 with the reference marked; the horizontal bar is the propagated uncertainty from the capacitance pair, not a confidence interval.

Table 9: Engineering benchmark comparison. The primary d_{HJ} column uses the Hammerstad–Jensen value in the denominator, as specified; the last column is the supplementary difference normalised by the FEM value and is not d_{HJ}.

Method	ε_{eff}	Z_0 (Ω)	$100 d_{\text{HJ}}$	Diff. vs FEM
FEM matched pair	3.3535	50.642	0.879%	0.000%
Hammerstad–Jensen	3.3008	50.201	0.000%	0.871%
IPC-2141	–	49.393	1.608%	2.466%

The Hammerstad–Jensen formula gives 50.201 Ω and IPC-2141 gives 49.393 Ω; the two approximations differ from each other by 0.81 Ω, which is seventeen times our own 0.048 Ω. That comparison is the useful one: the disagreement between standard formulas, not the agreement of any one of them with the FEM value, sets the scale at which an impedance can be claimed. The FEM result is also 0.642 Ω above the common 50 Ω design target, a difference of 1.28% that is design context rather than a numerical reference.

12 Limitations

The model is electrostatic and quasi-TEM. It omits dielectric and conductor loss, surface roughness, dispersion, radiation, solder mask and manufacturing tolerance; the conductors are perfect. Those omissions are larger than the 0.1% numerical question the report answers, and no numerical refinement removes them.

Three numerical limitations are specific to this work. The reference of (8) is an extrapolation from two families that have not reached their asymptotic regime: it is the best available limit rather than an exact value, and the agreement of the two routes bounds the extrapolation error without proving it small. The 0.18% domain figure measures the sensitivity of the answer to the domain choice, obtained on production meshes, and is not a measured truncation error of the M7 solution. The grading analysis establishes a constant-factor gain on one background family under one marking rule and nothing further; it does not establish optimality, and other rules were not tested.

The adaptive campaign of Section 8 is a limitation as well as a check. Its numbers are energy-norm numbers measured against frozen M6/M7 references, not capacitance numbers against the extrapolated limit, and the two experiments were not run at matched targets. It reduced the unknown count at the accuracy levels of interest but not the total runtime, which is why the submitted values rest on a single uniform mesh rather than on an adaptive sequence. That is a deliberate refusal, recorded rather than omitted.

13 Conclusions

On the studied cross-section the submitted values are $C = 120.620\,\mathrm{pF/m}$, $0.153\,\mathrm{pF/m}$ below the extrapolated limit, and $C_{\mathrm{air}} = 35.968\,\mathrm{pF/m}$ on the same M7 mesh, hence $\varepsilon_{\mathrm{eff}} = 3.3535$ and $Z_0 = 50.642\,\Omega$. These sit 0.127%, 0.07% and 0.09% away from limits obtained by two independent routes that agree to 2.9×10^{-6}.

Four findings underpin the numbers. 23 analytic configurations are reproduced to roundoff, including a separate audit of the Dirichlet row on non-zero boundary data. The usual convergence indicator understates the error by the factor $2^p - 1 = 1.23$, because the measured order is 1.16 rather than one. That order is set by the corner and not by the basis, since P_2 gives 1.15. And grading the corner reaches 0.017% with 139,426 unknowns, 7.6× closer than M7 at 39% of its size, with a plateau that falls fourfold per background level.

The remaining error is therefore understood rather than merely small: it is the discretisation of one point, it can be bought down by concentrating elements there, and the two other contributions — truncation at 0.18% and algebraic error thirteen orders below it — are quantified separately so that none of them is hidden inside a single quoted figure.

References

[1] Wolfgang Hackbusch. *Elliptic Differential Equations: Theory and Numerical Treatment*, volume 18 of *Springer Series in Computational Mathematics*. Springer, 2 edition, 2017.

[2] Ralf Hiptmair. Numerical methods for partial differential equations. Lecture notes, ETH Zurich, 2025. Spring Term 2025, course 401-0674-00L.

[3] Alfio Quarteroni. *Numerical Models for Differential Problems*, volume 16 of *MS&A – Modeling, Simulation and Applications*. Springer, 3 edition, 2017.

A AI Transparency Audit

We used AI/Codex for code scaffolding, plotting, workflow generation, documentation, consistency checks and report drafting and revision. Every numerical claim below was re-checked against stored output, and the checks are designed so that a wrong claim fails loudly rather than degrading gracefully. The first four cases record assumptions, possible falsifiers, evidence locations and decisions.

The audit follows the five-gate process required for AI-assisted solver work: formula transcription checked against the project documents; structure and indexing tested through geometry, material-region, boundary-node and degree-of-freedom checks; the formulation verified by deriving the weak form, the elimination and the capacitance identities explicitly; independent reproduction addressed by the manufactured solution, the layered problem, the two-route reference and the boundary-row audit; and fragile results interrogated through the domain, reference-stability, grading and tolerance studies rather than accepted at face value.

Table 10: Project-specific AI audit summary.

Component	Main risk	Verification evidence
P_1 assembly and elimination	Incorrect assembly, boundary treatment or coefficient handling	23 analytic configurations; manufactured solution; layered problem; boundary-row audit
Microstrip geometry	Trace hole, interface or boundary nodes misidentified	Geometry tests; full and near-trace figures; potential bounds and symmetry
Capacitance extraction	Reaction sign or trace-node summation error	C_E/C_Q agreement at 10^{-15}; charge-balance defect
Reference value	Extrapolation of a sequence that is not asymptotic	Two families with different elements and codes, agreeing to 2.9×10^{-6}
Order diagnosis	Attributing a reduced rate to the wrong cause	P_1 and P_2 slopes compared; corner exponent fitted independently
Corner grading	Confusing a constant-factor gain with a better rate	Four backgrounds; plateau ratios constant at about four
Domain study	Changing the far field also changes the near-trace mesh	Fixed near field; refined $12H$ repeat separates mesh from truncation
Final capacitance	Selecting a convenient mesh rather than an evidenced one	M3–M7 sequence; change-versus-error table; frozen mesh fingerprint
Matched all-air solve	Remeshing instead of changing only the material	Frozen mesh hash, same DOFs, same boundary sets, same sparsity pattern
Engineering benchmarks	Circular validation using FEM outputs inside the formulas	Benchmark functions accept only geometry and ε_r; formula spread quoted

Case audit 1: energy–charge agreement. *Claim examined:* near-equality of C_E and C_Q proves spatial convergence.
Assumption: both quantities use the same discrete potential, operator, prescribed voltage and element integrals.
Risk and falsifier: reactions and energy come from the same stiffness matrix, so even a coarse mesh satisfies $2W_e = V_0 Q_{\text{trace}}$.

Independent evidence: M3 has a C_E/C_Q difference of 1.0×10^{-15} but $C_E = 124.273\,\mathrm{pF/m}$, which is 3.16% from the limit.
Decision: reject the inference; use the energy–charge difference for algebraic consistency and the mesh sequence for spatial sensitivity.

Case audit 2: the finite far boundary. *Claim examined:* the tested $12H$ solution is the exact infinite-domain answer.
Risk and falsifier: a finite zero-flux boundary still perturbs the field, so one large box cannot remove truncation.
Independent evidence: the fixed-core $3H$–$12H$ sequence and the refined $12H$ repeat separate the two effects; the repeat moves only 0.036% while the $8H$ difference is 0.18%.
Decision: call $12H$ a tested finite-domain reference and report 0.18% as a sensitivity indicator, not a bound.

Case audit 3: errors across different meshes. *Claim examined:* coefficient vectors from different meshes can be subtracted elementwise.
Risk and falsifier: vector entries live in different bases, and a fine triangle crossing a coarse gradient jump invalidates constant-gradient integration.
Evidence: in the earlier graded comparison, 21440 of 180224 reference triangles crossed a coarse diagonal, about 4.55% of the field-domain area.
Decision: never subtract coefficients across meshes; transfer the coarse gradient onto the reference elements and integrate there. The grading results of Section 7 use this route, and the earlier centroid-based curves are retained only as history.

Case audit 4: reading a plateau. *Claim examined:* a graded error that stops falling has reached the discretisation floor.
Risk and falsifier: the plateau is set by the background mesh, not by roundoff, so the same campaign on a finer background should keep falling.
Evidence: four backgrounds give plateaus of 1.087%, 0.270%, 0.069% and 0.017%, in constant ratio 4.02–4.12.
Decision: report the grading gain as a constant factor tied to the background, never as an improved rate; state the background whenever the graded error is quoted.

Reproduction commands. The report and the slides share one number source. From the repository root:

```
python3 -m pip install -r requirements.txt
pytest -q
python3 presentation/make_figures.py   # figures + presentation/numbers.tex
python3 report/build_numbers.py        # report/numbers.tex
python3 report/make_field_figures.py   # field map and layered interface
python3 report/make_report_figures.py  # truncation and final-capacitance charts
python3 report/make_campaign_figures.py  # grading campaign, cost
python3 report/make_adaptive_figures.py  # adaptive meshes, three-way plot
cd report && latexmk -pdf main.tex     # or tectonic -X compile main.tex
python3 code_hardened/boundary_row_audit.py
```

Most figures are produced by `presentation/make_figures.py` and copied unchanged into `report/figures_v2/`, so a figure in the paper and the corresponding slide are the same file. The field map, the layered-interface panels, the grading campaign, the adaptive mesh snapshots and the three-way comparison are redrawn for the page by the four `report/make_*.py` scripts listed above, from the same stored solves and numbers: the adaptive panels read `results/adaptive_extension/meshes/A0,A3,A7.npz` and the cost panels read `results/adaptive_extension/three_way_comparison.csv`. Timings vary between runs; the numbers quoted here are those of the stored output.

B Individual Contribution Statements

Member 1:
Chuqiao Tang (2360692). **[TO BE COMPLETED BY TEAM.]**

Member 2:
Jinyu Ma (2363433). **[TO BE COMPLETED BY TEAM.]**

Member 3:
Muqing Xu (2362900). **[TO BE COMPLETED BY TEAM.]**

Member 4:
Shuoran Fu (2362626).

I worked on developing and organising the Python FEM code, including mesh generation, boundary conditions, sparse matrix assembly and solution of the electrostatic problem. My work covered files such as `code/geometry.py`, `code/fem_solver.py` and `code/run_all.py`. I also worked on the numerical experiments: the verification suite, the domain study, the uniform sequence and the reference value, and the corner-grading campaign analysed in Section 7.

For the report, I designed the structure and wrote the technical sections in `report/main.tex`, including the argument that separates the distance from the limit from the change between meshes, and the diagnostics that attribute the reduced order to the corner. I organised the figures and tables so that every number is produced by a script from the stored tables.

AI tools assisted with coding, debugging and writing. I reviewed the generated content and checked the numerical claims against the stored output. My main contribution was connecting the implementation, the numerical experiments and the report into one consistent argument.

Repository evidence: commits `2b9dfed`, `b5eee20` and `7119103` record the code, report and frozen-mesh uploads under Shuoran Fu's account; `c10eabf` and `ccd2f37` add the hardened verification suite and the boundary-row audit. These are delivery records, not a complete record of individual work.

Member 5:
Taiqi Zhang (2363684). **[TO BE COMPLETED BY TEAM.]**

Member 6:
Yujia Ding (2362094). **[TO BE COMPLETED BY TEAM.]**